IU International University of Applied Sciences

Habit Tracker App - Abstract / Making Of

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Course: Object Oriented and Functional Programming with Python (DLBDSOOFPP01)

GitHub: https://github.com/rajifaruq2021-hue/oofpp_habits_project

1. Project Summary

For this portfolio project I designed and built a Python-based Habit Tracker application from scratch. The goal was to build a clean, well-documented backend for a habit tracking system - one that supports creating habits, checking them off, tracking streaks over time, and analysing progress - all through a command-line interface and without any graphical user interface.

The final product is a fully working application consisting of five Python modules totalling over 2,300 lines of carefully commented code, a SQLite database for saved storage, 5 pre-defined habits loaded with four weeks of realistic example data, and a test suite of 20 unit tests that all pass successfully.

2. Design Decisions and Approach

I began by reading the assignment brief carefully and identifying the two main programming style of programmings required: Object-Oriented Programming (OOP) for building a model of the habit itself, and Functional Programming for the analytics layer.

The Habit class was the natural starting point. A habit is a real-world entity with its own identity, its own data, and its own behaviour. Encoding it as a class with attributes (name, periodicity, checkoffs) and methods (checkoff(), get_current_streak(), get_longest_streak()) made the code clean, reusable, and easy to understand - even for a beginner.

For data storage I chose SQLite over a simple JSON file because it is a real relational database, teaches important concepts (tables, foreign keys, SQL), and is already built into Python - no installation required. The database layer in db.py is deliberately kept separate from the Habit class, following the principle of keeping each part of the code in its own file.

The analytics module was written entirely in the functional style. I used Python's built-in filter(), map(), and max() functions to answer questions about habits. Each analytics function is pure - it takes data in and returns a result without changing the original data. This makes the analytics code easy to test and reason about.

The CLI menu in main.py was designed to be as simple as possible: the user just types a number and presses Enter. No complex commands or flags are needed. The menu loops continuously until the user chooses to exit.

3. Challenges and How I Solved Them

The most challenging part of the the code I wrote was the streak calculation logic. Calculating whether two periods (days or weeks) are consecutive required converting timestamps into ISO week labels for weekly habits, and then comparing Monday dates using Python's date.fromisocalendar() method. I solved this with a private helper method _are_consecutive() inside the Habit class, which handles both daily and weekly cases cleanly.

Designing realistic pre-defined test fixture data was also an important step. I made the data intentionally imperfect - with a few missed days and one missed week - so that the streak-break logic could be tested meaningfully and the analytics functions would return varied, interesting results.

4. What I Learned

This project gave me hands-on experience with both OOP and functional programming in real, working Python code. I learned how to design a class that models a real-world concept, how to use SQLite for data saving data, how to write pure functions for analytics, and how to write unit tests that give confidence in the correctness of complex logic. I also learned the importance of separating concerns - keeping the Habit class, the database layer, and the analytics layer in separate modules made the code far easier to maintain and test.

5. GitHub Repository

The full source code, including all Python files, the README with installation instructions, and the complete unit test suite, is publicly available at:

https://github.com/rajifaruq2021-hue/oofpp_habits_project

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